

Math 3IA3 - Assignment 3

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Contents

Exercise 3.4.6

Let $x_n := n^{\frac{1}{n}}$ for $n \in \mathbb{N}$.

- (a) Show that $x_{n+1} < x_n$ if and only if $(1 + \frac{1}{n})^n < n$, and infer that the inequality is valid for $n \geq 3$. Conclude that (x_n) is ultimately decreasing and that $x := \lim_{n \rightarrow \infty} (x_n)$ exists.

Solution:

- (I) Show that $x_{n+1} < x_n \iff (1 + \frac{1}{n})^n < n$.

$$\begin{aligned} x_{n+1} < x_n &\implies \\ (n+1)^{\frac{1}{n+1}} < n^{\frac{1}{n}} &\implies \\ \ln(n+1)^{\frac{1}{n+1}} < \ln n^{\frac{1}{n}} & \\ \frac{1}{n+1} \ln(n+1) < \frac{1}{n} \ln n & \\ \frac{n}{n+1} \ln(n+1) < \ln n & \\ (1 + \frac{1}{n}) \ln(n+1) < \ln n & \\ e^{(1+\frac{1}{n}) \ln(n+1)} < e^{\ln n} & \\ (1 + \frac{1}{n})^{1+\frac{1}{n}} < n & \end{aligned}$$

- (II) Show that $(1 + \frac{1}{n})^n < n$, $\forall n \geq 3$.

- (III) Show $x := \lim_{n \rightarrow \infty} (x_n)$ exists.

(b) Use the fact that the subsequence also converges to x to conclude that $x = 1$.

Exercise 3.5.5

If $x_n := \sqrt{n}$, show that (x_n) satisfies $\lim_{n \rightarrow \infty} |x_{n+1} - x_n| = 0$ but is not a Cauchy sequence.

Solution:

(I) Show that x_n satisfies $\lim_{n \rightarrow \infty} |x_{n+1} - x_n| = 0$.

$$x_n \geq 1 \quad \forall n \in \mathbb{N}$$

$$\begin{aligned} |(x_{n+1} - x_n) - 0| &= |x_{n+1} - x_n| \\ &= |\sqrt{n+1} - \sqrt{n}| \\ &= \sqrt{n+1} - \sqrt{n} \end{aligned}$$

(II) Show that x_n is not a Cauchy sequence.

We show that x_n is not a Cauchy sequence by proving that it diverges because, over the set of real numbers, a sequence is Cauchy if and only if it is convergent to a real number. Let $M^2 > 0$ be arbitrary. By the Archimedean property, we know that there exists an $N \in \mathbb{N}$ such that for all $n \geq N \implies n > M^2$. Thus:

$$\begin{aligned} n &\geq N \implies \\ n &> M^2 \implies \\ \sqrt{n} &> \sqrt{M^2} \implies \\ x_n &> M \end{aligned}$$

Therefore, we know that $\lim_{n \rightarrow \infty} (x_n)$ diverges to infinity and therefore (x_n) cannot be Cauchy.

Exercise 3.6.10

Show that if $\lim_{n \rightarrow \infty} \left(\frac{a_n}{n}\right) = L$, where $L > 0$, then $\lim_{n \rightarrow \infty} (a_n) = +\infty$.

Solution:

Let $M > 0$ be arbitrary. Since $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$ is known, we can conclude using algebraic limit rules that $\lim_{n \rightarrow \infty} \left(\frac{M}{n}\right) = 0$. Since $\lim_{n \rightarrow \infty} \left(\frac{a_n}{n}\right) = L > 0$, we know there exists some $N_0 \in \mathbb{N}$ such that $\forall n \geq N_0$:

$$|\frac{a_n}{n} - L| < \frac{L}{2}$$

And therefore, $\frac{L}{2} < \frac{a_n}{n} < \frac{3}{2}L$ for all such n . Likewise, we know that there exists some $N_1 \in \mathbb{N}$ such that $\forall n \geq N_1$:

$$|\frac{M}{n}| < \frac{L}{2}$$

And therefore, $0 < \frac{M}{n} < \frac{L}{2}$ for all such n . Thus, letting $N = \max\{N_0, N_1\}$, we have that $\forall n \geq N$:

$$\begin{aligned} \frac{a_n}{n} &> \frac{M}{n} \\ a_n &> M \end{aligned}$$

Therefore, since for arbitrary $M > 0$, we can always find an $N \in \mathbb{N}$ such that $a_n > M \forall n \geq N$, $\lim_{n \rightarrow \infty} (a_n) = +\infty$.

Exercise 3.7.11

If $\sum_{n=1}^{\infty} a_n$ with $a_n > 0$ is convergent, then is $\sum_{n=1}^{\infty} (a_n)^2$ always convergent? Either prove it or give a counter-example.

Solution:

Proof. We know, by definition of infinite sums, that if $\sum_{n=1}^{\infty} a_n$ is convergent, then the sequence of its partial sums $(s_k) = \sum_{n=1}^k (a_n)$ must converge to zero, and therefore, $\lim_{k \rightarrow \infty} (s_k) = 0$. By using algebraic limit rules, we can therefore find that:

$$\begin{aligned} \lim_{k \rightarrow \infty} \left(\sum_{k=1}^{\infty} (s_k)^2 \right) &= \lim_{k \rightarrow \infty} \left(\sum_{n=1}^k (a_n)^2 \right) \\ &= \sum_{n=1}^{\infty} (a_n)(a_n) \end{aligned}$$

□

Exercise 4.1.7

Show that $\lim_{x \rightarrow c} (x^3) = c^3$ for any $c \in \mathbb{R}$.

Solution:

We can easily show this using the sequential definition of functional limits. Let (x_n) be any sequence defined on $\mathbb{R}$ (since the domain of x^3 is $\mathbb{R}$) such that $x_n \neq c \forall n \in \mathbb{N}$ and $\lim_{n \rightarrow \infty} (x_n) = c$. By sequential limit theorems, we know that:

$$\begin{aligned} \lim_{n \rightarrow \infty} (x_n^3) &= (\lim_{n \rightarrow \infty} (x_n))^3 \\ &= c^3 \end{aligned}$$

Therefore, by definition of functional limits, since for all such (x_n) , $\lim_{n \rightarrow \infty} (x_n^3) = c^3$, $\lim_{x \rightarrow c} (x^3) = c^3$.